```
for each training step t:
  # Muon: every weight matrix W (n×m)
  M ← μ·M + G
  O ← NewtonSchulz(M) · 0.2·√max(n,m)
  W ← W − η·(O + λ·W)
  # QK-Clip: every attention head h
  if S_max[h] > τ:
    γ ← τ / S_max[h]
    W_qc ← √γ·W_qc;  W_kc ← √γ·W_kc
    W_qr ← γ·W_qr    # shared k_R untouched
```

update RMS matched
to AdamW: reuse its LR

reuses the forward's
per-head max logit

runs after the update
this step's pass unchanged